

Trial Examination 2024

## HSC Year 12 Mathematics Advanced

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### General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

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### Total Marks: 100

#### Section I – 10 marks (pages 2–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

#### Section II – 90 marks (pages 8–30)

- Attempt Questions 11–29
- Allow about 2 hours and 45 minutes for this section

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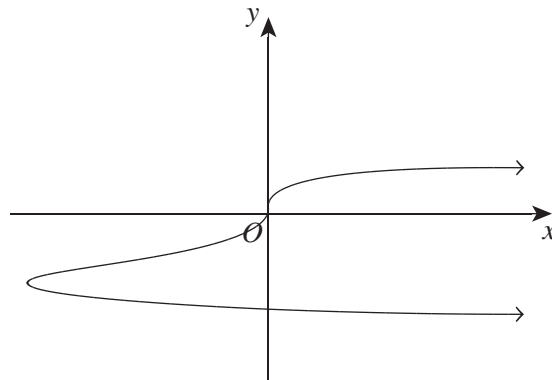
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**SECTION I****10 marks****Attempt Questions 1–10****Allow about 15 minutes for this section**Use the multiple-choice answer sheet for Questions 1–10.

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- 1 The graph of a relation is shown.



Which type of relationship does the graph represent?

- A. one-to-one
  - B. one-to-many
  - C. many-to-one
  - D. many-to-many
- 2 Using the trapezoidal rule with four intervals, what is the approximate area under the curve  $y = \sqrt{2x+1}$  between  $x = 0$  and  $x = 4$ ?
- A.  $2 + \sqrt{3} + \sqrt{5} + \sqrt{7}$
  - B.  $2 + 2(\sqrt{3} + \sqrt{5} + \sqrt{7})$
  - C.  $4 + 2(\sqrt{3} + \sqrt{5} + \sqrt{7})$
  - D.  $4 + \sqrt{3} + \sqrt{5} + \sqrt{7}$

- 3 Families in a particular suburb were surveyed about the types of pets they own. The results of the survey are shown.

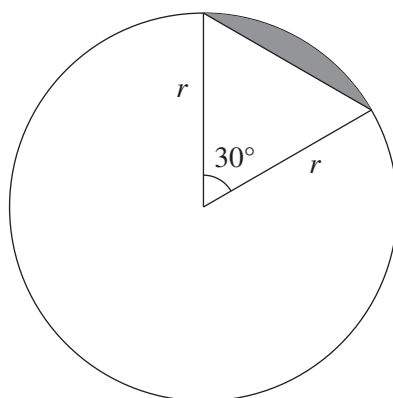
- 40% of families own a dog.
- 36% of families own a cat.
- 18% of families own both a cat and a dog.

A family is chosen at random and found to own a cat.

What is the probability that the family also owns a dog?

- A. 18%
- B. 36%
- C. 40%
- D. 50%
- 4 Which of the following is the domain of the function  $y = \frac{1}{\sqrt{x^2 - 1}}$ ?
- A.  $(-\infty, -1] \cup [1, \infty)$
- B.  $(-\infty, -1) \cup (1, \infty)$
- C.  $(-1, 1)$
- D.  $[-1, 1]$

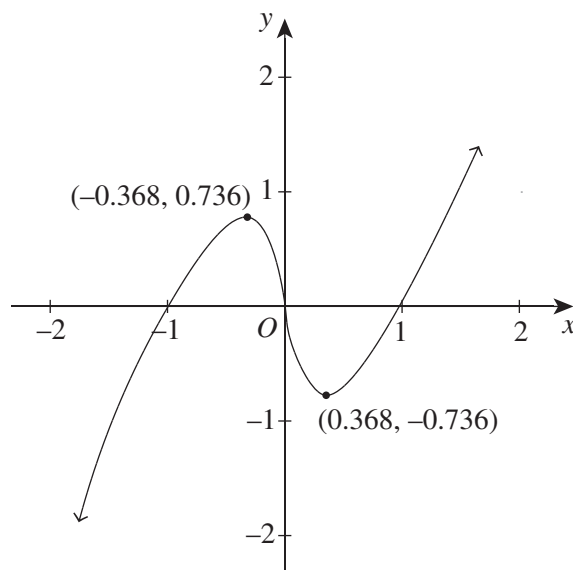
- 5 The diagram shows a circle with radius  $r$  cm and a sector with an internal angle of  $30^\circ$ .



If the area of the shaded segment is  $1.7 \text{ cm}^2$ , which of the following is closest to the radius of the circle?

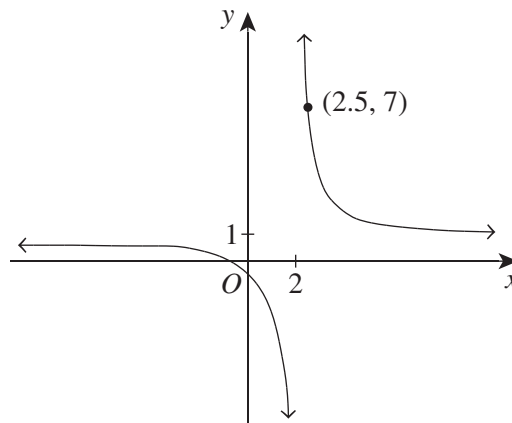
- A. 0.12 cm
- B. 0.34 cm
- C. 12.00 cm
- D. 144.08 cm

- 6 The graph of  $y = x \ln x^2$  is shown.



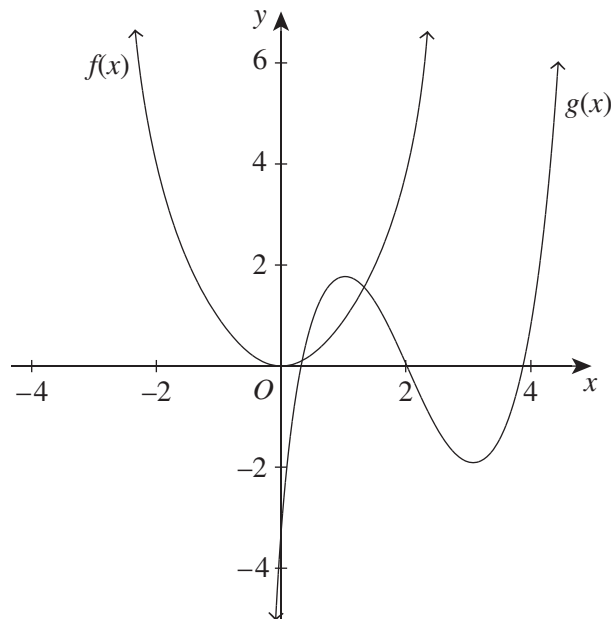
- How many solutions does the equation  $|x \ln x^2| = \frac{1}{3}(x+1)$  have?
- A. 3  
B. 4  
C. 5  
D. 6
- 7 A supermarket assistant sets up a display of lemonade cans. The display forms a triangle, where each row has one fewer can than the one below. There are 15 cans at the base of the display and one can at the top of the display.  
What is the total number of cans used in the display?
- A. 30  
B. 120  
C. 160  
D. 180
- 8 Alycia and Spyro are participating in a fitness program. The probability that Alycia will complete the program is 0.8, and the probability that Spyro will complete the program is 0.7.  
What is the probability that only one of them will successfully complete the program?
- A. 0.14  
B. 0.24  
C. 0.38  
D. 0.56

- 9 The graph of  $y = \frac{a}{x+h} + k$  is shown.



Which of the following statements is correct?

- A.  $h < k < a$
  - B.  $h < a < k$
  - C.  $k < h < a$
  - D.  $k < a < h$
- 10 The graphs of  $f(x) = x^2$  and  $y = g(x)$  are shown.



How many stationary points does the graph of  $y = g(f(x))$  have?

- A. 1
- B. 2
- C. 4
- D. 5

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# HSC Year 12 Mathematics Advanced

## Section II Answer Booklet

**90 marks**

**Attempt Questions 11–29**

**Allow about 2 hours and 45 minutes for this section**

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### Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
  - Your responses should include relevant mathematical reasoning and/or calculations.
  - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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**Please turn over**

**Question 11** (2 marks)

Find the derivative of the function  $y = xe^{7x}$ . Give your answer in factorised form.

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**Question 12** (3 marks)

A bag contains six cards that are each printed with one of the following numbers.

**3**

111, 121, 122, 211, 221, 222

A student chooses one card at random.

Let  $N_1$  be the event that the first digit of the number on the card is 1 and  $N_2$  be the event that the second digit of the number on the card is 1.

Determine whether  $N_1$  and  $N_2$  are independent events.

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**Question 13** (3 marks)

Solve the equation  $\frac{\operatorname{cosec}(x)}{5} - 2 = 0$  for  $0 \leq x \leq 360^\circ$ . Give your answer correct to the nearest minute. **3**

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**Question 14** (4 marks)

Alan works as a sales representative for a company that sells household appliances. The number of sales he makes in a day is given by the probability distribution table.

|            |      |      |      |      |      |      |
|------------|------|------|------|------|------|------|
| $x$        | 0    | 1    | 2    | 3    | 4    | 5    |
| $P(X = x)$ | 0.10 | 0.20 | 0.26 | 0.15 | 0.14 | 0.11 |

- (a) Find the expected value of the number of sales that Alan makes on a particular day. **1**

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- (b) Another sales representative, Lena, has an expected number of sales of 3.5 and a standard deviation in the number of sales of 1.2. **3**

The CEO of the company would like to promote a sales representative to be the team leader.

If the CEO bases their decision on sales performance, are they more likely to promote Alan or Lena? Justify your answer with appropriate statistical reasoning.

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**Question 15** (3 marks)Find the primitive function of  $7^x + \cos(x) + \sec^2(3x)$ .**3**

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**Question 16** (7 marks)

Greg is a student who conducted a study to investigate whether there was a correlation between the number of texts a person sends each day and the amount of reading they completed each day.

A group of seven people participated in the study. On a particular day, each person reported how many texts they had sent, and how many pages of a book they had read.

Greg released the results shown at the conclusion of the study.

- The number of texts sent by each participant was: 15, 20, 24, 25, 28, 28, 28.
- The total number of pages read by the participants was 98.
- If a participant sends 0 texts in a day, it is predicted that they would read 17 pages on that day.

Let  $x$  be the number of texts sent in a day and  $y$  be the number of pages of a book read in a day.

- (a) Greg predicts that the least-squares regression line would pass through the  $y$ -intercept and the point  $(\bar{x}, \bar{y})$ , where  $\bar{x}$  is the sample mean of the number of texts sent in a day and  $\bar{y}$  is the sample mean of the number of pages of a book read in a day. **3**

Find the equation of Greg's regression line.

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- (b) Use Greg's regression line to predict the number of texts sent by a person who has read six pages in a day. Comment on the accuracy of this prediction and the use of Greg's regression line. **2**

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**Question 16 continues on page 13**

## Question 16 (continued)

- (c) Viki, a fellow student, disagrees with Greg's results. She asks the same group of seven people to record the same data on a different day. The results of Viki's study are shown in the table.

**2**

|     |    |    |    |    |    |    |    |
|-----|----|----|----|----|----|----|----|
| $x$ | 11 | 19 | 25 | 26 | 28 | 28 | 31 |
| $y$ | 14 | 10 | 11 | 18 | 14 | 16 | 15 |

Using the data from Viki's study, determine the equation of the least-squares regression line. Comment on the accuracy of each student's equation.

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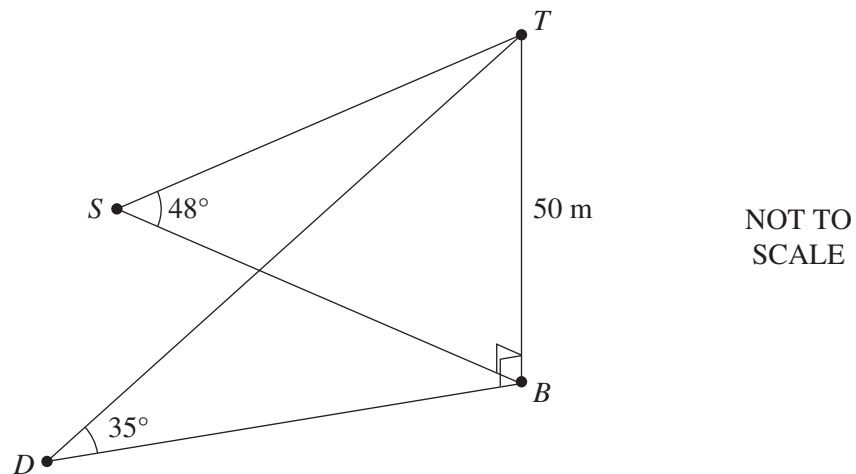
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**End of Question 16**

**Question 17** (5 marks)

Darline and Saher visit Sydney Harbour on New Year's Eve to watch the fireworks.

Darline is standing outside the Sydney Opera House and Saher is on a yacht in the harbour. They can both see the fireworks at the top of the Sydney Harbour Bridge.



From Darline's position ( $D$ ), the Sydney Harbour Bridge ( $B$ ) is at a bearing of  $78^\circ$ , and the angle of elevation to the top of the bridge ( $T$ ) is  $35^\circ$ .

From Saher's position ( $S$ ), the Sydney Harbour Bridge is at a bearing of  $140^\circ$ , and the angle of elevation to the top of the bridge is  $48^\circ$ .

The top of the bridge is 50 m above the base of the bridge.

- (a) Using appropriate diagrams and reasoning, show that  $\angle SBD = 62^\circ$ .

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**Question 17 continues on page 15**

## Question 17 (continued)

- (b) Find the distance between Saher and Darline. Give your answer correct to two decimal places. **3**

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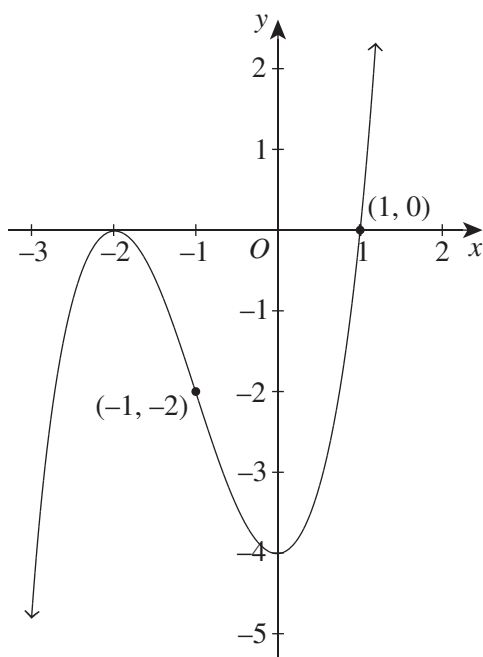
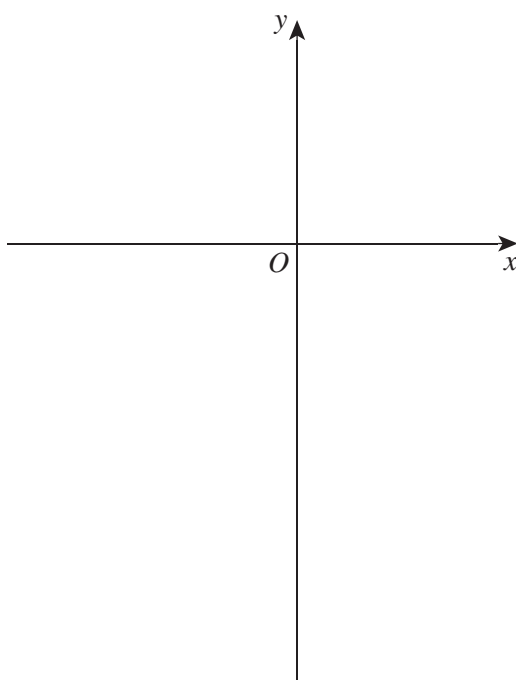
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**End of Question 17**

**Question 18** (3 marks)The diagram shows the graph of  $y = f(x)$ .**3**Sketch the graph of  $y = f(2x - 1)$ , labelling all stationary points and intercepts.



**Question 19** (3 marks)

- (a) Sketch the graph of  $y = \sqrt{25 - x^2}$ .

**1**

- (b) Hence, or otherwise, find  $\int_{-5}^5 \sqrt{25 - x^2} + 3 dx$ .

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**Question 20** (5 marks)

As part of a research study, a group of tadpoles is introduced into a large pond. The population of tadpoles in the pond,  $P$ , after  $t$  weeks is modelled by the equation

$$P = \frac{2500}{1 + 3^{-t}}.$$

- (a) Find the initial population of tadpoles in the pond. 1

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- (b) Show that  $\frac{dP}{dt} = \frac{2500 \ln 3}{3^t (1 + 3^{-t})^2}$ . 2

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- (c) Hence, or otherwise, determine whether the population of tadpoles is increasing or decreasing over time. Justify your answer with appropriate mathematical reasoning. 2

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**Question 21** (6 marks)

A company produces boxes that contain various miniature chocolates.

The chocolate boxes are labelled with a net weight of 200 g. However, the company aims to fill the chocolate boxes to a net weight of 205 g to avoid customer complaints.

The weight of the chocolate boxes is normally distributed with a mean of 205 g and a standard deviation of 2 g.

- (a) Approximately 30% of the boxes weigh less than 204 g.

**3**

What is the approximate percentage of chocolate boxes that weigh between 206 g and 207 g?

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- (b) The table gives the probability that a random variable that is normally distributed lies below  $z$  for some positive values of  $z$ .

**3**

| $z$ | 0       | 0.01    | 0.02    | 0.03    | 0.04    | 0.05    | 0.06    |
|-----|---------|---------|---------|---------|---------|---------|---------|
| 1.7 | 0.95543 | 0.95637 | 0.95728 | 0.95818 | 0.95907 | 0.95994 | 0.96080 |
| 1.8 | 0.96407 | 0.96485 | 0.96562 | 0.96638 | 0.96712 | 0.96784 | 0.96856 |
| 1.9 | 0.97128 | 0.97193 | 0.97257 | 0.97320 | 0.97381 | 0.97441 | 0.97500 |
| 2   | 0.97725 | 0.97778 | 0.97831 | 0.97882 | 0.97932 | 0.97982 | 0.98030 |

Find the probability that a chocolate box weighs more than 208.6 g.

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**Question 22** (4 marks)

- (a) Show that  $\frac{1}{\sec \theta - \tan \theta} - \frac{1}{\sec \theta + \tan \theta} = 2 \tan \theta$ . 2

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- (b) Hence, or otherwise, find  $\int \frac{1}{\sec \theta - \tan \theta} - \frac{1}{\sec \theta + \tan \theta} d\theta$ . 2

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**Question 23** (8 marks)

The function  $f(x)$  has an  $x$ -intercept at  $(2, 0)$ .

The derivative of the function is  $f'(x) = 4(x - 1)^3 - 9(x - 1)^2$ .

- (a) Show that  $f(x) = (x - 1)^4 - 3(x - 1)^3 + 2$ . **2**

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- (b) Find the coordinates of any stationary point(s) of  $f(x)$  and determine their nature. **4**

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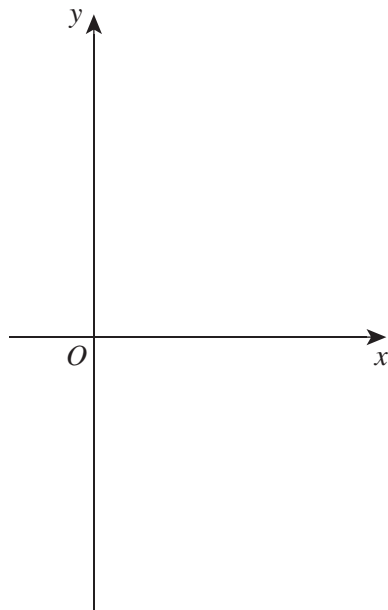
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**Question 23 continues on page 22**

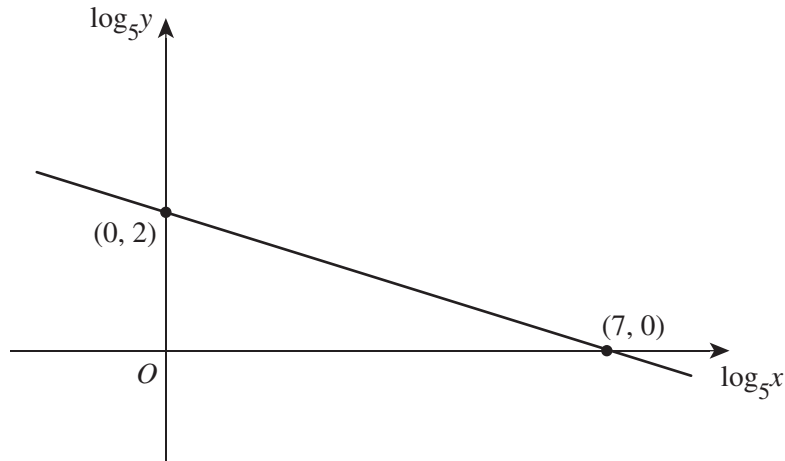
## Question 23 (continued)

- (c) Sketch the graph of  $y = f(x)$ , labelling the y-intercept and all points found in part (b).

**2****End of Question 23**

**Question 24** (3 marks)

The graph shows the linear relationship between  $\log_5 x$  and  $\log_5 y$ .

**3**

The Cartesian equation for the relationship between  $x$  and  $y$  has the form  $x^a y^b = c$ .  
Find the values of  $a$ ,  $b$  and  $c$ . Write the value of  $c$  using index notation.

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**Question 25** (6 marks)

A particle moves along a straight line. The velocity of the particle,  $v$ , in  $\text{m s}^{-1}$ , is given by

$$v(t) = 1 - 3 \cos\left(\frac{\pi t}{5}\right),$$

where  $t$  is the time, in seconds, and  $0 \leq t \leq 10$ .

- (a) Find the particle's maximum and minimum velocity.

**1**

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- (b) Hence, or otherwise, sketch the graph of the particle's velocity, labelling all important features.

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- (c) Determine how many times the particle changes its direction. Explain your answer with reference to the graph in part (b).

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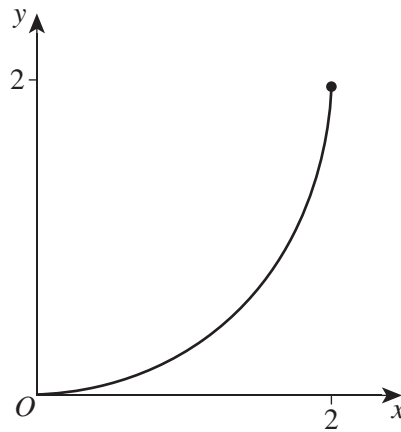
**Question 26** (6 marks)

A continuous random variable has the probability density function  $f(x)$  given by

$$f(x) = \begin{cases} \frac{kx}{5-x^2} & 0 \leq x \leq 2 \\ 0 & \text{for all other values of } x \end{cases}$$

where  $k$  is a positive constant.

The graph of  $y = f(x)$  is shown.



- (a) Show that the value of  $k$  is  $\frac{2}{\ln 5}$ .

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**Question 26 continues on page 26**

## Question 26 (continued)

- (b) Show that the cumulative distribution function  $F(x)$  is given by

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$$F(x) = \begin{cases} 0 & x < 0 \\ 1 - \log_5(5 - x^2) & 0 \leq x \leq 2 \\ 1 & x > 2 \end{cases}$$

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- (c) Hence, or otherwise, show that the median of  $F(x)$  is  $\sqrt{5 - \sqrt{5}}$ .

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**End of Question 26**

**Question 27** (6 marks)

Amy decides to invest money in a bank account to start saving for a house deposit. She invests an initial amount of \$1000. At the beginning of each month, Amy invests an amount that is 0.5% more than the amount invested in the previous month. At the end of each month, the bank pays interest of 0.4% per month, compounding monthly.

Let  $A_n$  be the balance at the end of the  $n$ th month.

- (a) Show that the balance of the account at the end of the second month is given by 2  
 $1000(1.004)^2 + 1000(1.005)(1.004)$ .

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- (b) Show that  $A_n = 1\,004\,000(1.005^n - 1.004^n)$ . 4

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**Question 28** (6 marks)

Consider the series  $\log_e x - 3(\log_e x)^2 + 9(\log_e x)^3 - 27(\log_e x)^4 + \dots$ .

- (a) Show that this series is a geometric series.

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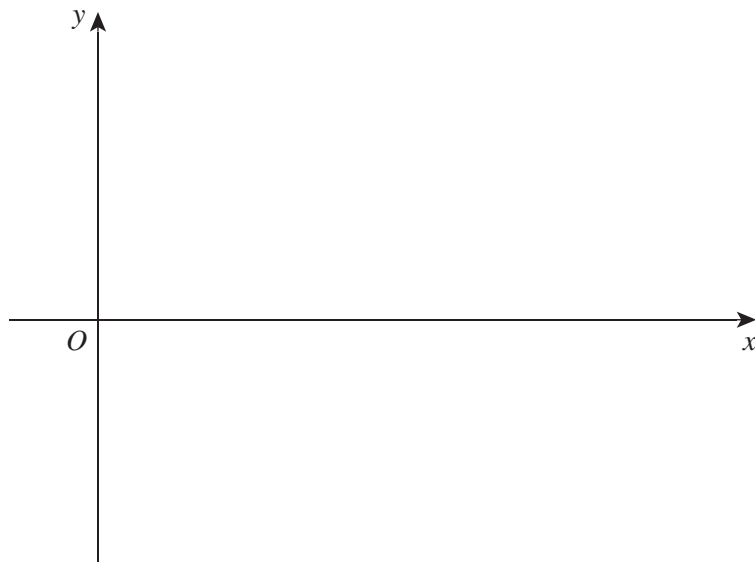
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- (b) On the same set of axes, sketch the graphs of  $y = \ln x$ ,  $y = -\frac{1}{3}$  and  $y = \frac{1}{3}$ . Show all intercepts.

**2**

- (c) Hence, or otherwise, find the values of  $x$  for which the series has a limiting sum.

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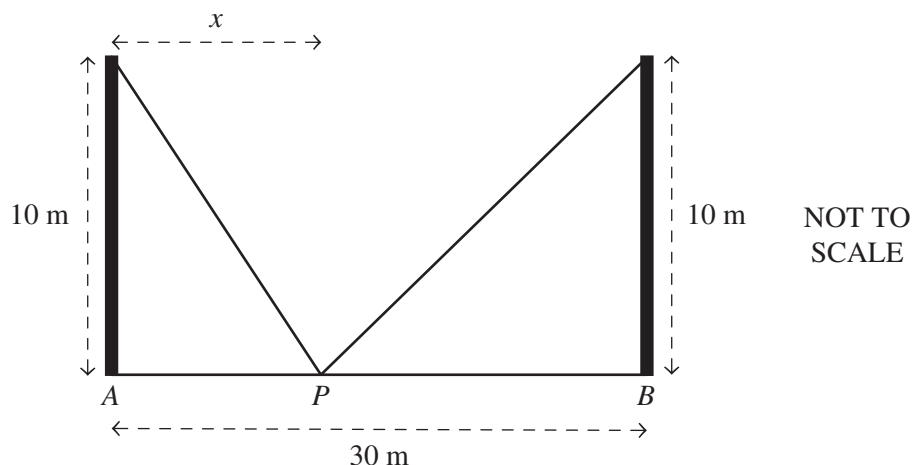
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**Question 29** (7 marks)

The diagram shows two 10 m poles,  $A$  and  $B$ , that are 30 metres apart. The ends of a single length of wire are attached to the top of each pole. The length of wire is then staked to the ground between them at a movable point,  $P$ .

Let  $x$  be the distance between  $A$  and  $P$ .



- (a) Show that the length of the wire,  $L$ , is given by  $L = \sqrt{x^2 + 100} + \sqrt{x^2 - 60x + 1000}$ .

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- (b) Show that  $\frac{dL}{dx} = \frac{x}{\sqrt{x^2 + 100}} + \frac{x - 30}{\sqrt{x^2 - 60x + 1000}}$ .

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**Question 29 continues on page 30**

Question 29 (continued)

- (c) Find the value of  $x$  when the minimum length of wire is used.3

End of paper



## Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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**MATHEMATICS ADVANCED**  
**MATHEMATICS EXTENSION 1**  
**MATHEMATICS EXTENSION 2**  
**REFERENCE SHEET**

**Measurement****Length**

$$l = \frac{\theta}{360} \times 2\pi r$$

**Area**

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

**Surface area**

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

**Volume**

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

**Functions**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For  $ax^3 + bx^2 + cx + d = 0$ :

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

**Relations**

$$(x - h)^2 + (y - k)^2 = r^2$$

**Financial Mathematics**

$$A = P(1 + r)^n$$

**Sequences and series**

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

**Logarithmic and Exponential Functions**

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

**Trigonometric Functions**

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

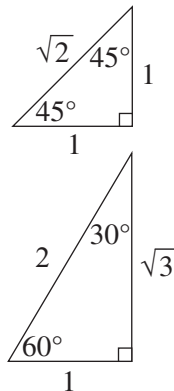
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$

**Trigonometric identities**

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

**Compound angles**

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$$

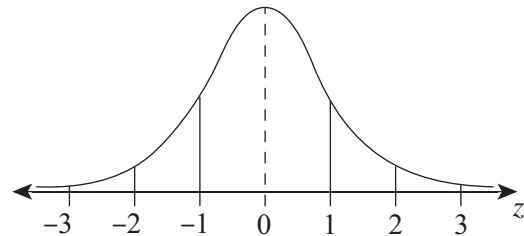
$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

**Statistical Analysis**

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score  
less than  $Q_1 - 1.5 \times IQR$   
or  
more than  $Q_3 + 1.5 \times IQR$

**Normal distribution**

- approximately 68% of scores have  $z$ -scores between  $-1$  and  $1$
- approximately 95% of scores have  $z$ -scores between  $-2$  and  $2$
- approximately 99.7% of scores have  $z$ -scores between  $-3$  and  $3$

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

**Probability**

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

**Continuous random variables**

$$P(X \leq r) = \int_a^r f(x)dx$$

$$P(a < X < b) = \int_a^b f(x)dx$$

**Binomial distribution**

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

**Differential Calculus****Function****Derivative**

$$y = f(x)^n$$

$$\frac{dy}{dx} = nf'(x)[f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \quad \text{where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

**Integral Calculus**

$$\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + c$$

where  $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$$

where  $a = x_0$  and  $b = x_n$

**Combinatorics**

$${}^n P_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^n C_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \dots + \binom{n}{r}x^{n-r}a^r + \dots + a^n$$

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**Vectors**

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}||\underline{v}|\cos\theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

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**Complex Numbers**

$$\begin{aligned} z = a + ib &= r(\cos\theta + i\sin\theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos\theta + i\sin\theta)]^n &= r^n(\cos n\theta + i\sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

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**Mechanics**

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$